PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE A · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

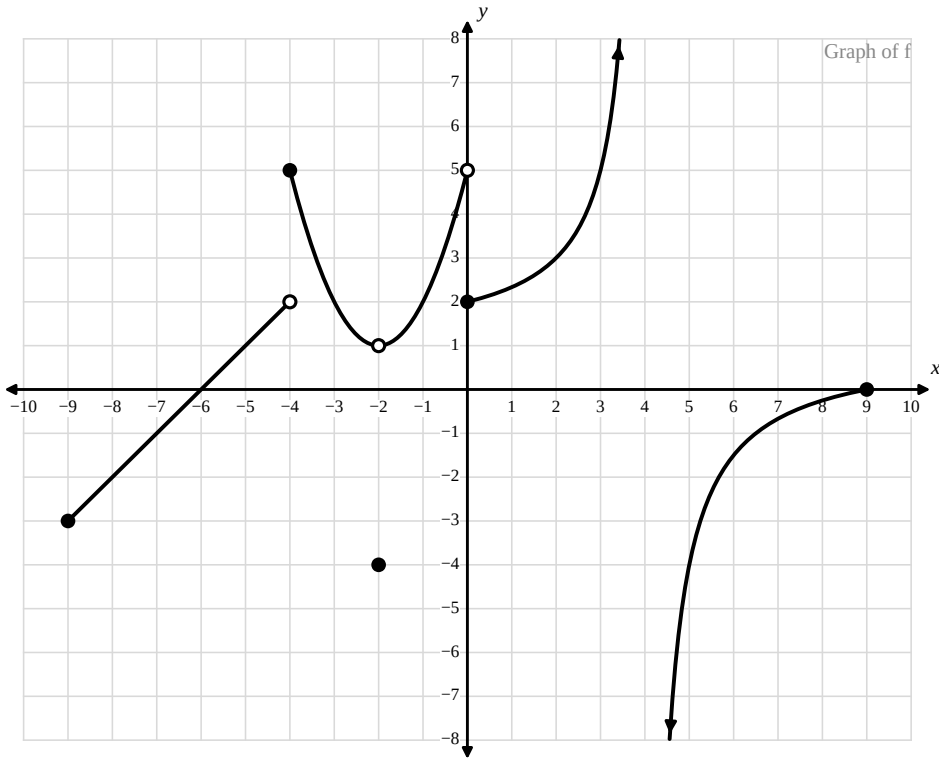
_____

_____

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

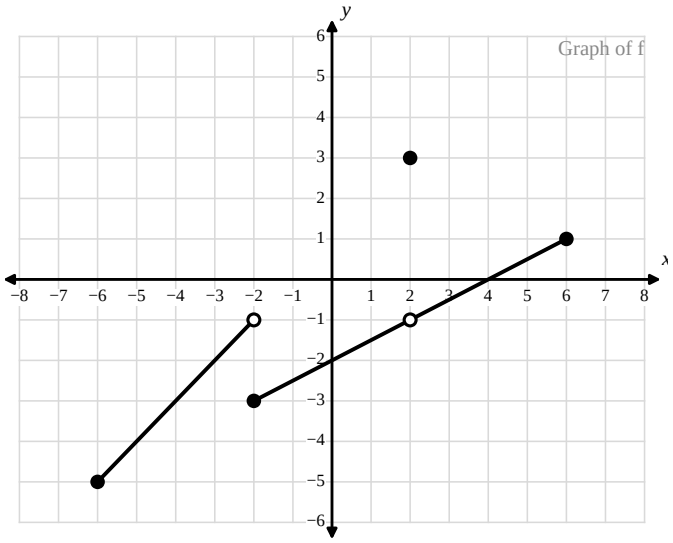


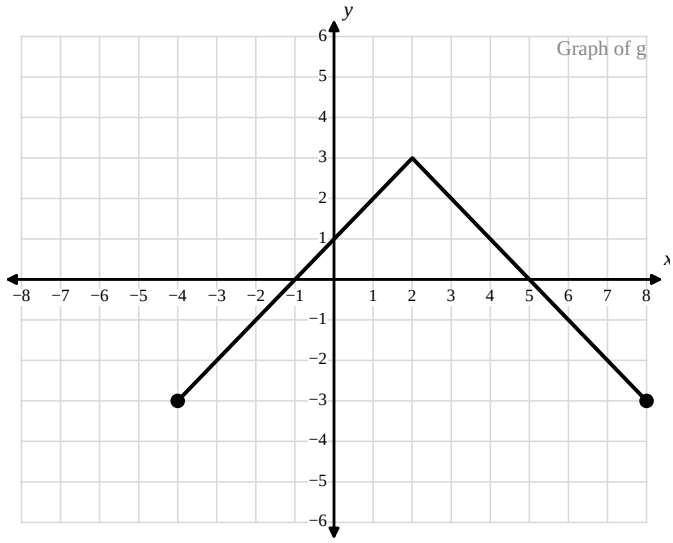
1. $\lim_{x\to -2} f(x) =$	**2.** $\lim_{x\to -4} f(x) =$	**3.** $\lim_{x\to 4^+} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 2$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x \to -2^+} \dfrac{[f(x)]^2 - 3g(x)}{g(x) + 2}$

(2 points)

6. Use the table to estimate $\lim\limits_{x \to 3} g(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	2.9	2.99	2.999	$\to 3 \leftarrow$	3.001	3.01	3.1
$g(x)$	0.2515823	0.2501564	0.2500156		0.2499844	0.2498439	0.2484567

$\lim\limits_{x \to 3^-} g(x) \approx$ _____ $\lim\limits_{x \to 3^+} g(x) \approx$ _____ $\lim\limits_{x \to 3} g(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x \to 3^-} \dfrac{x^2 - 9}{|x - 3|}$

8. $\lim\limits_{x \to -3} \dfrac{x^3 + 27}{x^2 + x - 6}$

9. $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{\sqrt{x + 3} - 2}$

10. $\lim\limits_{x \to 2} \dfrac{\frac{1}{x^2} - \frac{1}{4}}{x - 2}$

11. For $f(x) = \dfrac{3}{x + 1}$, find $\lim\limits_{h \to 0} \dfrac{f(x + h) - f(x)}{h}$

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

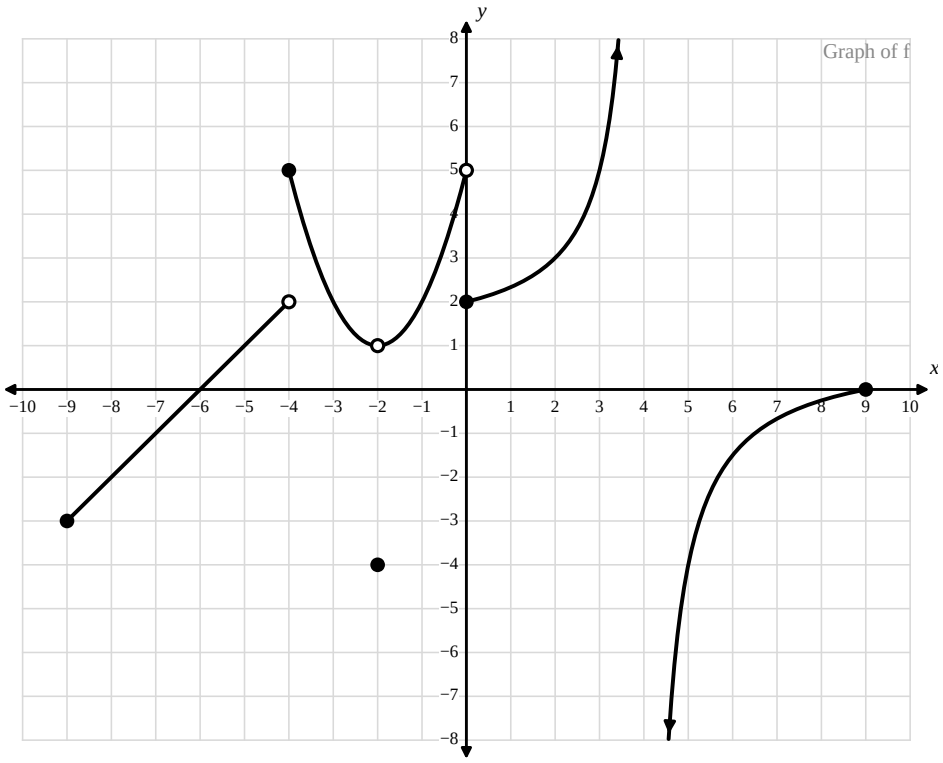
PRACTICE A · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to -2} f(x) =$

Both sides of the curve head to the open circle at $(-2, 1)$.

1

2. $\lim_{x\to -4} f(x) =$

$\lim_{x\to -4^-} f(x) = 2$, $\lim_{x\to -4^+} f(x) = 5$

$2 \neq 5$, *so* $\lim_{x\to -4} f(x)$ *does not exist.*

DNE

3. $\lim_{x\to 4^+} f(x) =$

Just right of $x = 4$ *the curve falls without bound.*

$-\infty$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 2$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

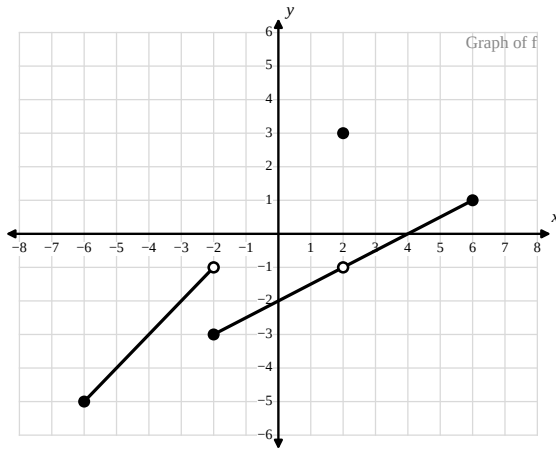
The curve reaches height 2 from both sides at $x = -3$ *and* $x = -1$.

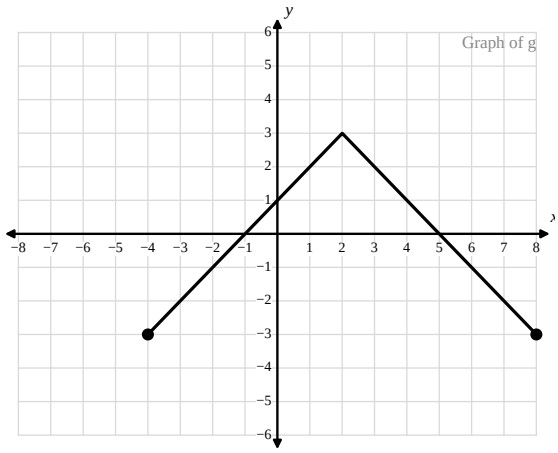
Not $x = -4$*: only the left side heads to 2 (the right side starts at 5).*

Not $x = 0$*: only the right side starts at 2 (the left side heads to 5).*

$a = -3,\ a = -1$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x\to -2^+} \dfrac{[f(x)]^2 - 3g(x)}{g(x)+2}$

$$\lim_{x\to -2^+} \frac{[f(x)]^2 - 3g(x)}{g(x)+2} = \frac{[\lim_{x\to -2^+} f(x)]^2 - 3\lim_{x\to -2^+} g(x)}{\lim_{x\to -2^+} g(x) + \lim_{x\to -2^+} 2}$$

Quotient (bottom limit is 1, not 0), power, difference, constant multiple.

$$= \frac{(-3)^2 - 3(-1)}{-1+2} = \frac{12}{1} = 12$$

12

(2 points)

6. Use the table to estimate $\lim\limits_{x\to 3} g(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	2.9	2.99	2.999	$\to 3 \leftarrow$	3.001	3.01	3.1
$g(x)$	0.2515823	0.2501564	0.2500156		0.2499844	0.2498439	0.2484567

Left: 0.2515823, 0.2501564, 0.2500156

Right: 0.2484567, 0.2498439, 0.2499844

Both sides agree. A table estimates; it does not prove.

≈ 0.25

$\lim\limits_{x\to 3^-} g(x) \approx$ 0.25 $\lim\limits_{x\to 3^+} g(x) \approx$ 0.25 $\lim\limits_{x\to 3} g(x) \approx$ 0.25

Evaluate each limit. (2 points each)

7. $\lim_{x\to 3^-} \dfrac{x^2-9}{|x-3|}$

Substitution produces 0/0: indeterminate.

For $x < 3$, $|x-3| = -(x-3)$.

$$\lim_{x\to 3^-} \frac{(x-3)(x+3)}{-(x-3)}$$
$$= \lim_{x\to 3^-} [-(x+3)]$$
$$= -(3+3) = -6$$

-6

8. $\lim_{x\to -3} \dfrac{x^3+27}{x^2+x-6}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to -3} \frac{(x+3)\left(x^2-3x+9\right)}{(x+3)(x-2)}$$
$$= \lim_{x\to -3} \frac{x^2-3x+9}{x-2}$$
$$= \frac{9+9+9}{-3-2} = -\frac{27}{5}$$

$-\frac{27}{5}$

9. $\lim_{x\to 1} \dfrac{x^2-1}{\sqrt{x+3}-2}$

Substitution produces 0/0: indeterminate.

$$\lim_{x\to 1} \frac{\left(x^2-1\right)\left(\sqrt{x+3}+2\right)}{\left(\sqrt{x+3}-2\right)\left(\sqrt{x+3}+2\right)}$$
$$= \lim_{x\to 1} \frac{(x-1)(x+1)\left(\sqrt{x+3}+2\right)}{(x+3)-4}$$
$$= \lim_{x\to 1} \frac{(x-1)(x+1)\left(\sqrt{x+3}+2\right)}{x-1}$$
$$= \lim_{x\to 1} (x+1)\left(\sqrt{x+3}+2\right)$$
$$= (2)(2+2) = 8$$

8

10. $\lim_{x\to 2} \dfrac{\frac{1}{x^2}-\frac{1}{4}}{x-2}$

Substitution produces 0/0: indeterminate.

LCD: $4x^2$

$$\lim_{x\to 2} \frac{\left(\frac{1}{x^2}-\frac{1}{4}\right)4x^2}{(x-2)\,4x^2}$$
$$= \lim_{x\to 2} \frac{4-x^2}{4x^2(x-2)}$$
$$= \lim_{x\to 2} \frac{-(x-2)(x+2)}{4x^2(x-2)}$$
$$= \lim_{x\to 2} \frac{-(x+2)}{4x^2} = \frac{-4}{16} = -\frac{1}{4}$$

$-\frac{1}{4}$

11. For $f(x) = \dfrac{3}{x+1}$, find $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$

$$\lim_{h\to 0} \frac{\frac{3}{x+h+1}-\frac{3}{x+1}}{h}$$

LCD: $(x+h+1)(x+1)$

$$= \lim_{h\to 0} \frac{3(x+1)-3(x+h+1)}{h(x+h+1)(x+1)}$$
$$= \lim_{h\to 0} \frac{-3h}{h(x+h+1)(x+1)}$$
$$= \lim_{h\to 0} \frac{-3}{(x+h+1)(x+1)} = -\frac{3}{(x+1)^2}$$

$-\frac{3}{(x+1)^2}$